Algorithm Design - First Midterm
15-11-2024, *Time: 120 minutes*
Prof. Aris Anagnostopoulos, Prof. Stefano Leonardi, Dr. Simone Fioravanti

Instructions:

- You are given the question sheet and 4 sheets for your answers. The problems are 4 (turn the page) but the 4th is *bonus*, meaning it gives you the possibility of 5 extra points.
- Before you start, write on the top of each sheet (1) your full name, (2) your student ID (*matricola*) and (3) the sheet number (1, 2, 3, and 4).
- If you use additional sheets as rough draft (*brutta*) for calculations, etc., you must hand it, along with the rest of the exam. You can keep the question sheet.
- Hand in the sheets in order **without attaching the sheets with each other.**
- When your pages are full, you can ask for extra pages; make sure that you also write there your name, student id, and the sheet number (5, 6, etc.). You must hand all the sheets that you receive.
- The rooms are small so it may be tempting to copy. Note that if we catch you copying in any way (copying from another student, using your phone, etc.) you will be automatically disqualified.

Problem 1

(10 points)

Given three sequences of characters $x[1,\dots,n]$, $y[1,\dots,m]$ and $z[1,\dots,l]$, we consider the problem of finding the longest common subsequence formed by contiguous or non contiguous characters that appear in the same order in all sequences x, y, z. For example, the longest common subsequence of sequences $x =$ abdefghm, $y =$ acdffhlm and $z =$ aggdefehm is $LCS(x, y, z) =$ adfhm.

1. Describe a dynamic program that finds the value in number of characters of the longest common subsequence of three strings of n, m and l characters.
2. Provide the pseudocode of the algorihm that finds the value of the optimal solution and the pseudocode of the algorithm that finds the longest common subsequence.
3. Prove that time and space complexity is $O(n \cdot m \cdot l)$.

Problem 2

(10 points)

A cloud computing company runs a set of n servers to be activated when rented from its clients. Server $i \in [n]$ produces a revenue $r(i)$ when it is activated. In order to activate server i, it is also required the activation of a customized dedicated infrastructure $j(i) \in [m]$ that has has cost $c(j)$ to be activated. Notice that a customized dedicated infrastructure can host more than 1 server, e.g., for servers i_1, i_2, i_3, $j(i_1) = j(i_2) = j(i_3) = j$. Each server i is therefore associated to an infrastructure $j(i)$ that must be activated if server i is activated.

1. Design a polynomial time algorithm that finds a set of servers and a set of infrastures to be activated in order to maximize the total profit, i.e., revenue minus cost.
2. Prove the correctness of the algorithm.
3. Study the computational complexity of the algorithm.

Problem 3

(10 points)

An edge e of an s–t network G is called *critical* if decreasing its capacity $c(e)$ results in a decrease in the maximum flow value F. Assume $F > 0$.

1. Prove that any edge crossing a minimum cut is critical.
2. Now, assume $c(e) \in \{0, 1\}$ for all $e \in E$ and let $k \leq F$ be a fixed integer. Suppose you want to decrease the max-flow value F as much as possible deleting at most k edges. Show an algorithm that does that.

3. Study the time complexity of the algorithm.

Problem 4

Bonus (+ 5 points)

You have n homework assignments due right now, but you haven't started any of them, so they are all going to be late. Each assignment requires d_i days to complete, and has a cost penalty of c_i per day. So if assignment i ends up being finished t days late, then it incurs a penalty of $t \cdot c_i$. Assume that once you start working on one assignment, you must work on it until you finish it, and that you cannot work on multiple assignments at the same time.

Example. You have $n = 3$ assignments: A_1 takes $d_1 = 2$ days and has a penalty of $c_1 = 10$ points/day, A_2 takes $d_2 = 3$ days and has a penalty of $c_2 = 30$ points/day, and A_3 takes $d_3 = 2$ days and has a penalty of $c_3 = 5$ points/day. The best order is then A_2, A_1, A_3 which results in a total penalty of $3 \cdot 30 + 10 \cdot (3+2) + 5 \cdot (3+2+2) = 175$ points.

1. Give an optimal **greedy algorithm** that outputs an ordering of the assignments that minimizes the total penalty (i.e. the sum of all penalties).
2. Prove the correctness of the algorithm.
3. Discuss the running time of the algorithm.

Hint. To get some intuition, try to compare all the possible solutions for the special cases of $n = 2$ and of $n = 3$.